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Full Length Research Paper

Determination of welding parameters for shielded metal arc welding

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Arc welding is a type of welding that uses a welding power supply to create an electric arc between an electrode and the base material to melt the metals at the welding point. They can use either direct (DC) or alternating (AC) current, and consumable or non-consumable electrodes. The process of arc welding is widely used because of its low capital and running costs. In this paper, arc welding parameters are determined for shielded metal arc welding method. Parameters are selected depending on welding methods and in the present study some parameters are included and some of them are excluded. To determine the welding parameters, the national and international welding standards and also welding experience are taken into consideration for shielded metal arc welding method. In the present study, the determined welding parameters are examined in detailed for shielded metal arc welding method and the resultant ideal welding parameters and their limits are given in equations and tables.

Key words: Shielded metal, arc welding, electric arc, welding parameters, optimization.

INTRODUCTION

Shielded metal arc welding (SMAW) method is a fusion-welding process in which coalescence of the metals is achieved by the heat from an electric arc between an electrode and the work (Groover, 2007; Soy, 2011). This process is one of the most employed processes of fabricating metal components. It is fundamentally a fusion of two or more pieces of metals by the application of heat and sometimes pressure (Agarwal, 1992). Thus welding involves a wide range of scientific variables such as time, temperature, electrode, power input and welding speed (Jariyabon et al., 2007; Lothongkum et al., 1999; Lothongkum et al., 2001; Karadeniz et al., 2007).

In this method, welding parameters are the most important factors affecting the quality, productivity and cost of welding joint (Karadeniz et al., 2007). Accordingly, these parameters affecting the arc and welding bath should be estimated and their changing conditions during process must be known before in order to obtain optimum results (Uslu et al., 2010; Ankara and Ozturk, 1997). The welding quality of the shielded metal arc welding is determined by the welding parameters including the welding slot forms, electrode diameter, welding current, welding speed, arc length, electrode advance angle, electrode oscillation angle and movement, welding direction and position, etc (Afolabi, 2008; Chan et al., 1999; Ates and Turker, 1999). In an effort to obtain high quality welds in shielded metal arc welding method, selection of ideal parameters should be performed according to engineering facts. Commonly, welding parameters are determined by trial and error, based on handbook values, and manufacturers' recommendations. On the other hand, this option may not yield optimal or in the vicinity of optimal welding performance (Karaoglu and Secgin, 2008).

In shielded metal arc welding method, the electrodes

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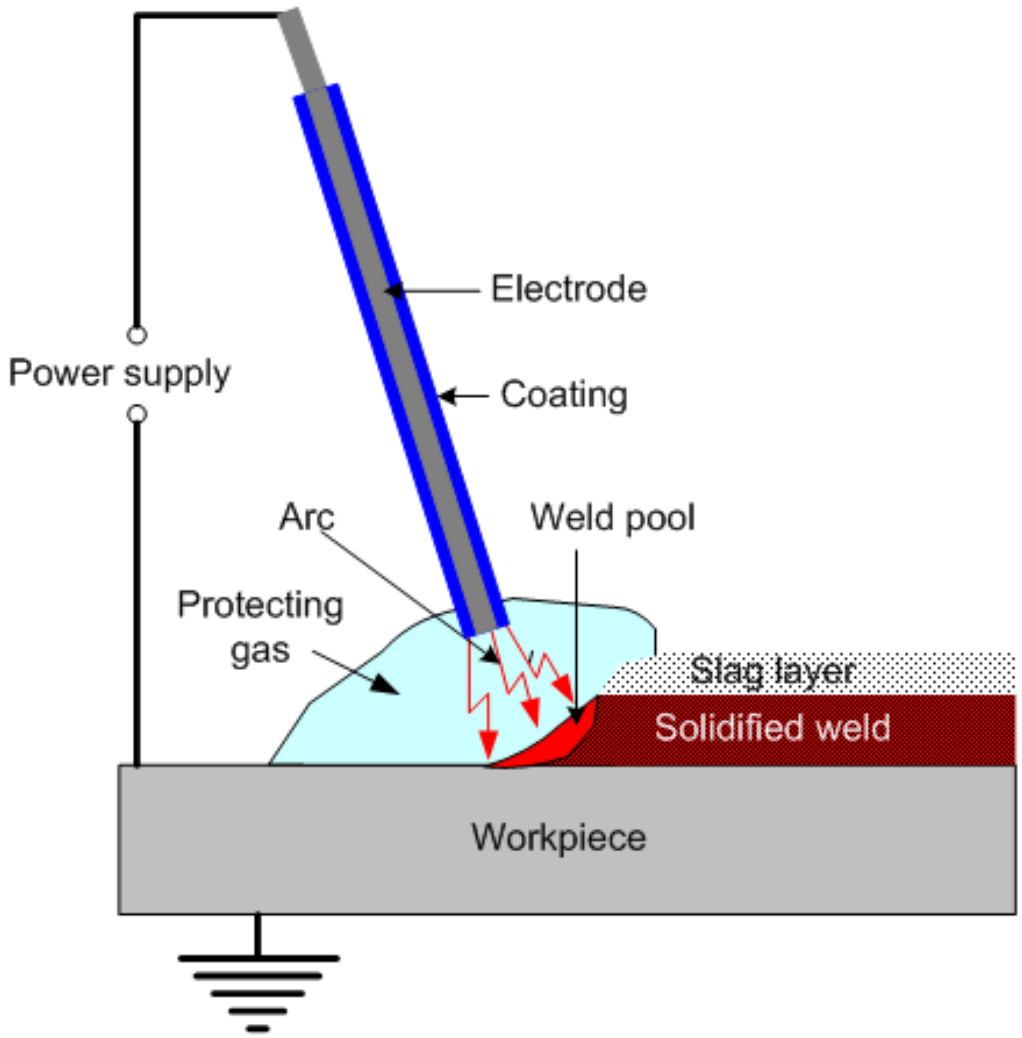


Figure 1. Schematic image of shielded metal arc welding.

are coated with a shielding flux of a suitable composition. The flux melts together with the electrode metallic core, forming a gas and a slag, shielding the arc and the weld pool. The flux cleans the metal surface, supplies some alloying elements to the weld, protects the molten metal from oxidation and stabilizes the arc. The slag is removed after Solidification. Figure 1 (Kopeliovich, 2010) shows a schematic image of shielded metal arc welding. As seen in this figure, process consists of electrode and its coating, arc formation, protecting gas, weld pool and solidified weld.

The main objective of this research was to determine arc parameters for shielded metal arc welding method. To determine the welding parameters, the national and international welding standards and also welding experience in application are taken into consideration for shielded metal arc welding method. In the present study, the determined welding parameters are examined in detailed for shielded metal arc welding method and the resultant ideal welding parameters and their limits are given in equations and tables.

DETERMINING OF SHIELDED METAL ARC WELDING PARAMETERS

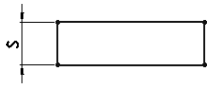
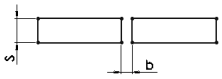
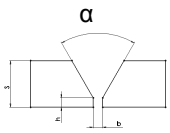
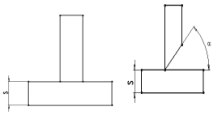
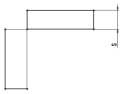
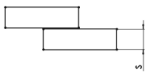
Materials type

Welding application will be realised for three different materials; namely plain carbon steel, alloy steel and stainless steel. Aluminium is not recommended for shielded metal arc welding method; therefore it is excluded in this technique.

Welding slot forms

Joining methods in welding design can be divided into four groups; such as butt, inner corner, outer corner and overlap. In welded parts, welding slot must be prepared to have better performance in

Table 1. Determination of dimensions in welding slot forms.

Code	Slot form	Schematic parade	α	b	Explanation
KA(1)	Plain weld		0°	0	
KA(2)	Butt weld		0°	If s ≤ 3, b = 0,5 mm If 3 < s ≤ 8, b = 2 mm If 8 < s ≤ 20, b = 3 mm If s > 20, b = 4 mm	If s>8 mm, butt weld is not recommended, v weld slot is suggested.
KA(3)	V weld		60°	If s ≤ 8, b = 0 mm If s > 8, b = 2 mm	If s ≥ 8, h = 3 mm (h = base height) If s < 8, v weld slot is not suggested, butt weld slot is recommended.
KA(4)	Inner corner		0° 60°	If s ≤ 8, b = 0 mm If s > 8, b = 0 mm	
KA(5)	Outer corner		0°	0	
KA(6)	Overlap		0°	0	

s = part thickness, b = bottom distance between two parts, α = Ideal angle value for V weld slot, KA (1) [weld slot (1)] = plain weld.

joined area. The types of welding slots are determined in the related standards. In Europe, EN ISO 9692-1:2003 international welding standard is valid in this subject and the types of welding slots are determined in detailed there.

The widely used welding slots are given in Table 1 according to the standard of EN ISO 9692-1:2003; in this table welding slot form and the distance (b) between work pieces to be welded, and the limit values to calculate this distance are given depending on the thickness of material (s).

Selection of electrode diameter

The electrodes used in shielded metal arc welding are divided in two main groups as joining and filler welding ones according to the purpose of the welding. The coated electrodes are also classified by the tensile strength of the deposited weld metal, the welding position in which they may used, the preferred type of current and polarity, and the type of coating. The metal wire used in the process is usually from 1.5 to 6.5 mm in diameter and 20 and 45 cm in length (Black and Kohser, 2008). The electrode material in welding is desired to be high strength, ductile and tough (Oguz, 1993). The molten electrodes provide both forming of arc and filling the welding area. According to EN ISO 2560:2005 standard, the electrodes are determined in the welding for plain carbon and low alloy steels. For selection of electrode; material type, welding position, welding current, welding slot form and work piece thickness above all are taken into consideration. The electrode diameter changes according to the material thickness and welding slot form. The most used electrodes in shielded metal arc welding applications are 2.50, 3.25 and 4.00 mm core diameter ones (Oz et al., 2011). The values of electrode core diameters are determined in Table 2 depending on work piece thickness.

Determining welding current

During the welding, that is, while arc occurs in welding period, current against working voltage is called as welding current. Welding machine is plugged into the alternative current and poles are determined. The cable tips connecting to electrode pliers and ground one are prepared, then electrode is attached to the pliers and arc occurs when electrode touches to work piece and consequently a permanent current circle continues.

Welding current is set by welders prior to welding application. During the welding application, the value of welding current is not

Table 2. Electrode core diameters suggested by program according to work piece thickness.

Work piece thickness (s)	Electrode core diameter (d)	Unit
S ≤ 3	2.5	mm
3 < S ≤ 20	3.25	mm
S > 20	4.00	mm

Table 3. Welding speeds according to work piece thickness (s), weld current (I) and electrode diameter (d).

Work piece thickness (S)	Welding speed (V_k), mm/s	Welding current (I)
S ≤ 3	4.50	d × 40 ampere
S > 3 ≤ 8	4.00	d×40 ampere
S > 8	3.50	d × 40 ampere

Table 4. The values of ideal advance angle and tolerances depending on welding position.

Welding position	Advance angle	Tolerance (º)
Plain weld	80	±5
Cornice (overlap) weld	80	±5
Vertical weld	105	±5
Overhead weld	80	±5

changed. However, arc is cut or current can be increased depending on welding application (Yasar et al., 2010). Welding current is selected as 40 folds of electrode core diameter (I = d × 40). This value can be changed as 10% depending on thickness of materials and position.

Welding speed

The movement of arc welding along work piece or the length of weld seam made in unit time is known as welding speed. When the speed is slow during the welding process, stock weld metal increases in the unit length and eventually it causes to enlarge the welding pool. With growing of weld metal and increasing of heat input, the molten metal flows into the front of arc within the welding slot and it affects the regular arc formation. The increment of speed causes to reduction of welding heat given to unit length and consequently the molten quantity of main metal decreases, this negatively affects the wetting of weld seam. The determined welding speeds are given in Table 3 according to the thickness of work piece (s), welding current (I) and diameter of electrode (d) (Durgutlu, 1997; Erturk, 1994).

Arc length

The importance of distance between electrode and work piece is vital for arc occurrence. The mentioning of arc length in various welding applications is required to understand the difference between arc lengths. If arc length is equal to electrode diameter, this is called as normal arc length. Long arc is obtained whenever arc length is greater than electrode diameter. The distances less than the electrode diameter are called as short arc length.

Experience shows that arc blowing is more effective in case of long arc length comparing to short arc one. For this reason, short arc length is always recommended for the work. It is also observed from the previous experience that, arc blowing will be less comparing to the uncoated or cored ones in case of welding with coated electrodes. In addition, blowing effect is more in thin coated electrodes comparing to thick ones (Anik, 1991; Anik and Tulbentci, 2007).

Electrode advance angle

The molten metal can be transferred by arc along the welding process and the welder should orientate the arc to form melting on joining surfaces. The angle between electrode and advance direction is generally 45 to 70°, however this value can also be changed between 45to 90°. The main principle here is the angle should prevent the flowing of slag in front of arc excluding vertical welding from top to down. The values of ideal advance angle and tolerances are given in Table 4 depending on welding position. The advance angle values are given according to advance direction. The schematic view of advance angles is illustrated in Figure 2 depending on welding position. Welding seam form changes in case of lowering tolerances and linear curves are extremely decreases, and burning grooves are seen in borders and penetration is reduced. Similar alteration in seam welding is obtained if tolerances are passed over, and arc blowing is realised (below 50°) and damage in borders is formed.

Electrode oscillation angle and movement

The parameter of oscillation angle directly affects the surface width of weld seam. If oscillation angle rises, the width of seam increases, when it decreases seam width lowers. Width changes depending on electrode core diameter (Welding seam width = d × 2.5 tolerance 20%). The equation oscillation angle is given in Equation

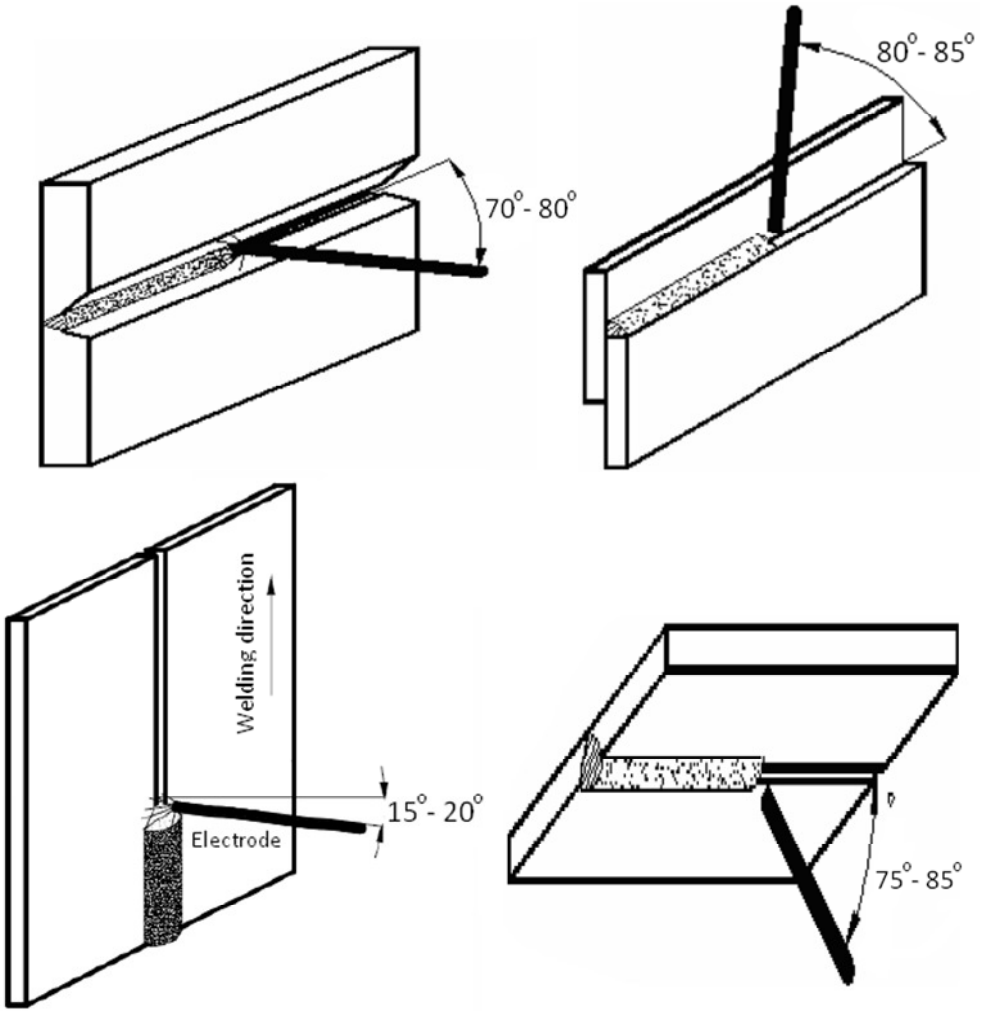


Figure 2. The schematic view of advance angles depending on welding position.

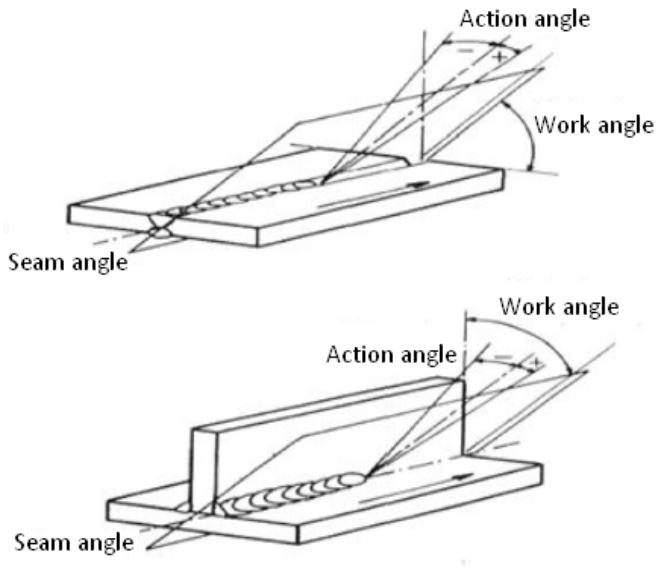


Figure 3. The schematic view of oscillation angle.

(2) and the schematic view of oscillation angle is shown in Figure 3. Oscillation (action) angle is symbolised as "β".

$$\beta = 5° \pm 2.5 \quad (2)$$

A plain welding seam is obtained if electrode moves in horizontal position, but if we move the electrode in various shapes, welding seam displays changeability. The important point here is keeping the hand movements of electrode as much as same of each other. This can be obtained as a result of developing the hand experience after for a while welding practice. While performing a plain seam, a simple pass action is done despite the electrode is generally depended on work piece thickness. Electrode is moved steadily without doing any action to right or left. The other electron action is called as zigzag (half moon). In this case, the electrode is equally moved to right and left obtaining welding seam.

Calculation of welding volume

Welding volume is the melting amount in 1 s in shielded metal arc welding process (Oguz, 2007). The formulation to calculate the welding volume is given in Equation 3. The electrode length molten in 1 s depending on electrode core diameter given in Table 5.

Table 5. The molten electrode length in 1 s depending on electrode core diameter.

Electrode core diameter (mm)	Molten electrode in 1 s (mm)
2.50	5.7
3.25	3.8
4.00	2.5

Table 6. Comparison of weld slots.

Specifications weld slot	α angle	Weld name	Suggestions
1	0	Plain weld	-----
2	0	Butt weld	If s>8 mm, butt weld is not recommended
3	60	V weld	If s<8 mm, v weld is not suggested
4	0	Inner corner weld	s ≤ 8 mm
	60	Inner corner weld	s > 8 mm
5	0	Outer corner weld	-----
6	0	Overlap weld	-----

$$H = \frac{\pi \times d^2}{4} \times h \times t \quad (3)$$

The parameters given in Equation 3 are defined as follows:

d= electrode diameter, h = electrode length molten in 1 s (mm), H = welding volume obtained in 1 s
t = time (s), V= advance velocity mm/s (note: welding speed).

The molten amount of electrode changes depending on welding current (that is I = d × 40). The given grade in Table 5 is average values.

RESULTS

In the present time, there are more than 100.000 materials spectrum used in industry. However, investigation of such a whole materials would take a lot of time, only some steels (plain carbon, alloy and stainless steels) and aluminium from the light materials are selected in the present work. By developing technology, it is possible to join aluminium by shielded metal arc welding as steel as possible. However, it is preferred to join aluminium only by MIG/MAG and TIG welding techniques due to obtaining better performance, spending shorter time and having economical production in these methods.

In the present investigation, 2.5, 3.25 and 4.00 mm diametric electrodes are selected. In fact, there are some other electrodes are produced under and over the selected diametric electrodes. However, the selected electrodes are mostly used in industry; therefore, they are especially selected for the present study.

It is known that, there are more than 10 forms used in welding slots in various international welding standards such as ISO EN, DIN, TS and AWS. However, in this study only six welding slot forms (Table 1) are used due to their most preference in industry. In Table 6, six welding slots were tabulated and compared in α angles and some recommendations are given accordingly with the type of welds.

Welding current can be changed depending on the thickness of work piece. If the used current is not satisfactory level, the electrode sticks to the work piece and weak welding is obtained depending upon the work piece thickness. In the other case, when the current is over the required level, some melting and distortions are formed on work piece. Evaluating the literature and the experimental findings, welding current span depending on the work piece, and the obtained welding performance are summarized in Table 7.

Arc disrupts in case of arc length moves away, then blowing rises up to the arc disruption, also the altitude of weld metal (h) gradually decreases when blowing increases and the width of weld seam rises. The voice of explosion and splash increases. In case of arc length comes nearer, the electrode sticks to work piece, then the weld seam seems to be swollen and welding is not realised if the electrode sticks to work piece. The alteration in arc length is given in Table 8 as moving away and coming nearer.

DISCUSSION

In this study, optimum welding current for shielded metal arc welding method values were determined. There is an

Table 7. Welding current (I) values depending on thickness of work piece.

Work piece thickness (S)	Welding current (I)	State of welding ≤
S ≤ 3	0 < I < 60	Electrode will stick to material
	60 < I < 90	Ideal welding
	90 < I < 350	Melting and distortion will gradually occur in material
3 < S ≤ 8	0 < I ≤ 60	Electrode will stick to material
	60 < I ≤ 80	Insufficient penetration
	80 < I ≤ 110	Ideal welding
	110 < I < 350	Melting and distortion will gradually occur in material
8 < S < 20	0 < I ≤ 60	Electrode will stick to material
	60 < I ≤ 90	Insufficient penetration
	90 < I ≤ 130	Ideal welding
	130 < I < 350	Melting and distortion will gradually occur in material
S > 20	0 < I ≤ 80	Electrode will stick to material
	80 < I ≤ 120	Insufficient penetration
	120 < I ≤ 150	Ideal welding
	150 < I < 350	Melting and distortion will gradually occur in material

Table 8. Variations in arc length moves away and comes nearer.

Variations	In case of arc length moves away	In case of arc length comes nearer
Arc length (a)	d ×1,25	d × 0,25
Blowing	Increases	Reduces
Weld metal height	Reduces	Increases
Welding seam	Increases	Reduces
Explosion and splash	Increases	Reduces
Light intensity	Increases	Reduces

optimum value of the welding current intensity (can be seen in Table 7). If this value is passed over, some deformations can be occurred in materials such as perforation, distortion, contraction etc. In addition, extreme brilliance and flush can also be observed in arc area. Consequently, the arc length rises uncontrollably. If current intensity is under optimum value, the required melting will not be formed and electrode can be stuck on materials, or very short arc length arises. The limit values related to determination of welding current depending on the thickness of work piece.

It is seen from this study that α angle, which is ideal angle value for weld slot, is generally required for 0° for plain, butt, overlap and outer corner welding. However, v type weld is needed 60° α angle. Only the inner corner can capable to have both 0 and 60° angles depending on the part thickness (s) values in case below or over than the 8 mm thickness. It is also seen that if s is greater than 8 mm, instead of butt weld, v weld is recommended and also if s is less than 8 mm butt welds is recommended as a substitute of v weld.

On the other hand, when arc length comes nearer to work piece or moves away, it is observed some alterations in welding seam, height of welding metal and light intensity such as explosion and splash. In case of moving away in arc length from the work piece, it is observed blowing, explosion and splash, and as a result intensity of light increases but welding metal height reduces. If the arc length comes to nearer to work piece; blowing, explosion and splash occur in welding seam and intensity of the light decreases and the height of welding metal increases. It is generally recommended that arc length is taken as half of the electrode core diameter and arc length is denoted by (a) letter.

Welding speed changes depending on the thickness of work piece. In this study, the suggested welding speed is about 3.5 mm/s in case of bigger than the 8 mm thick part. In case of increasing the welding speed, both the volume of welding metal and the heat input decrease in unit distance. On the other hand, in case of decreasing the welding speed, welding metal volume and also heat input increase in unit space. In case of using unsuitable

welding speed, an excess heat input to the work piece occurs and as a result some distortion and burning form.

In case of increasing in electrode oscillation angle, the width of welding seam raises; if the angle decreases the size of welding seam reduces. Furthermore, welding seam width shows a variation depending on electrode core diameter. In welding application, electrode movement affects the form of welding seam. It is important for welding performance in advancing the electrode movement along the welding line without deviating from the neutral axis and the controlling of electrode in equal frequency as far as possible. As a result of the completed studies, it is determined that the advance angle showed a variation comparing to welding position. It is recommended that the advance angle is 80° in plain, cornice (overlap) and overhead welding, but it is 105° in vertical welding to obtain maximum performance.

Conclusions

In the present work, arc parameters are studied and optimum welding parameters area determined for shielded metal arc welding method mostly used in arc welding techniques in industry. The determined welding parameters are used in a computer program to develop a welding simulator for selected welding technique, materials thickness and welding position. These parameters will appear on the screen as tables and then the screen will be printed and these results are given to welding trainees during the education and then welding process will be realised. As a result, taking into account of the determined parameters by software, the welding will be realised and the ideal welding will be obtained.

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